

For the expression, the first step to completing the square is shown.

$$x^2 - 8x + 15$$

$$(x^2 - 8x) + 15$$

- Complete the perfect square trinomial.
- In the expression, the value that was added to create the perfect square trinomial is being multiplied by _____, so the value that should be subtracted from 15 to maintain equivalency is _____.

Complete the table to complete the square without using algebra tiles.

	Process	Equation
	Original equation	$y = -8x^2 - 16x + 42$
	Factor out the common factor of the first two terms.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x) + 42$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 42$
4.	Then, to maintain equivalence, multiply the factor by the value. Add the opposite of that value.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 42$ + _____
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 + 42 + \underline{\hspace{1cm}}$
6.	Combine like terms.	$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 10x - 24$

8. Key Features

Key Features					
x -intercept		vertex		domain	
x -intercept		increasing		range	
y -intercept		decreasing		opens	
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$			

9. $y = 6x^2 + 12x - 48$

10. Key Features

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x -intercept		increasing		range	
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